

Numerical Integration

→ Numerical Integration is the process to calculate the value of definite integration

$$\int_a^b f(u) \cdot du$$

- Some method for solving numerical integration.

i) Trapezoidal Rule

ii) Simpson's $\frac{1}{3}$ Rule

iii) Simpson's $\frac{3}{8}$ Rule

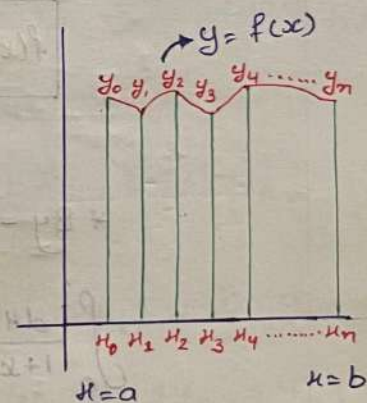
Trapezoidal Rule (Theory)

* Explain trapezoidal rule for solving definite integration (Numerical Integration)

→ Now we divide the graph of $f(u)$ into (n) number of small trapezium with height of is trapezium small

$$h = \frac{(b-a)}{n} \text{ where,}$$

n = no of sub interval



$$\int_a^b f(u) du = \text{Area}(\text{trap}(u_0, y_0, u_1, y_1)) + \text{Area}(\text{trap}(u_1, y_1, u_2, y_2)) + \text{Area}(\text{trap}(u_2, y_2, u_3, y_3)) + \dots + \text{Area}(\text{trap}(u_{n-1}, y_{n-1}, u_n, y_n))$$

$$= \frac{h}{2} (y_0 + y_1) + \frac{h}{2} (y_1 + y_2) + \frac{h}{2} (y_2 + y_3) + \dots + \frac{h}{2} (y_{n-1} + y_n)$$

$$= \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

where, " $h = \frac{b-a}{n}$ "

(Practical)

* Find the definite integration by using trapezoidal rule :-

Soln

$$\int_0^1 \frac{dx}{1+x^2}$$

Given that:

$$f(x) = \frac{1}{1+x^2}$$

$$a=0, \quad b=1, \quad n=4$$

Let,

$$h = \frac{b-a}{n} = \frac{1-0}{4} = 1/4$$

$$h = 1/4$$

x	0	$1/4$	$1/2$	$3/4$	1
$f(x)$	1	$16/17$	$4/5$	$16/25$	$1/2$
	y_0	y_1	y_2	y_3	y_4

* By - Trapezoidal:

$$\int_0^1 \frac{dx}{1+x^2} = \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)]$$

$$= \frac{1}{4 \cdot 2} [(1 + 1/2) + 2(\frac{16}{17} + \frac{4}{5} + \frac{16}{25})]$$

$$= \frac{1}{8} (\frac{3}{2} + \frac{32}{17} + \frac{8}{5} + \frac{32}{25})$$

$$= \frac{1}{8} (6.2623)$$

$$= 0.7827$$

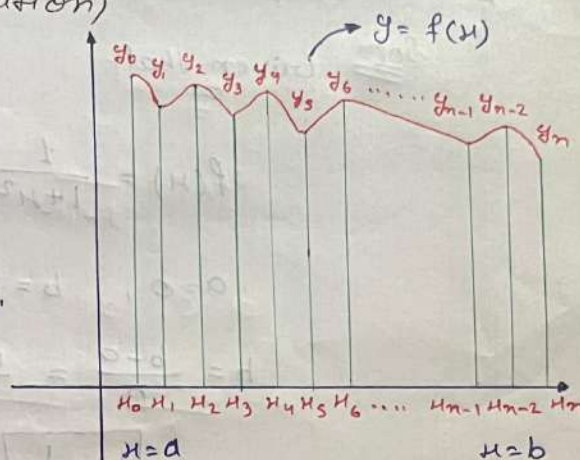
Simpson's $\frac{1}{3}$ Rule

(Theory)

* Explain Simpson's $\frac{1}{3}$ Rule for definite integration (Numerical integration)

Let, $f(x)$ be the real value continuous function define on close interval $[a, b]$ be one to find.

$$\int_a^b f(x) dx$$



we divide the graph of $f(x)$ n (no) of partition of parabola where, n is no corresponding to each pair of sub interval

$$\int_a^b f(x) dx = \text{Area}(\text{para}(x_0, y_0, y_1, y_2, x_2)) + \text{Area}(\text{para}(x_2, y_2, y_3, y_4, x_4)) + \dots + \text{Area}(\text{para}(x_{n-2}, y_{n-2}, y_{n-1}, y_n, x_n))$$

$$= \frac{h}{3} (y_0 + 4y_1 + y_2) + \frac{h}{3} (y_2 + 4y_3 + y_4) + \frac{h}{3} (y_4 + 4y_5 + y_6) + \dots + \frac{h}{3} (y_{n-2} + 4y_{n-1} + y_n)$$

$$= \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + y_8 + \dots + y_{n-2}) \right]$$

(Practical)

* Evaluate the definite integration by using Simpson's $\frac{1}{3}$ rule:

Soln Given that:-

$$\int_0^1 \frac{dx}{1+x^2}$$

$$f(x) = \frac{1}{1+x^2}$$

$$a=0, \quad b=1, \quad n=6$$

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
$f(x)$	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

* By - Simpson's $\frac{1}{3}$:

$$\int_0^1 \frac{1}{1+x^2} dx = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3 \times 6} \left[\left(1 + \frac{1}{2}\right) + 4\left(\frac{36}{37} + \frac{4}{5} + \frac{36}{61}\right) + 2\left(\frac{9}{10} + \frac{9}{13}\right) \right]$$

$$= \frac{1}{18} \left(\frac{3}{2} + \frac{144}{37} + \frac{16}{5} + \frac{144}{61} + \frac{18}{10} + \frac{18}{13} \right)$$

$$= \frac{1}{18} (\dots)$$

$$= 0.7853$$

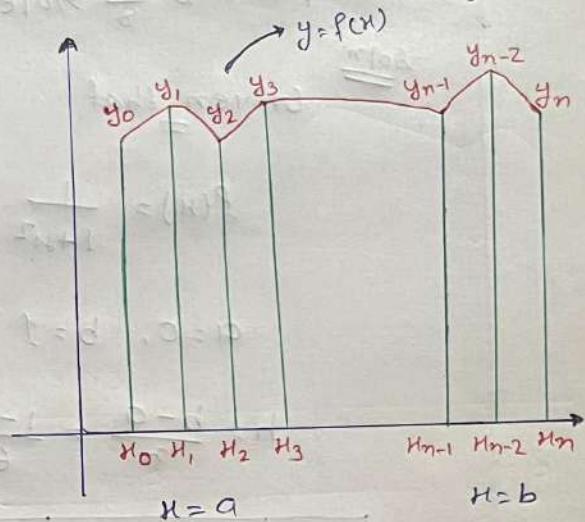
Simpson's $\frac{3}{8}$ Rule (Theory)

* Explain Simpson's $\frac{3}{8}$ rule for definite integration (numerical integration)

Let, $f(x)$ be the real valued continuous function defined on close interval

$[a, b]$ be one to find

$$\int_a^b f(x) dx$$



We divide the graph of $f(x)$ into n (no) of partitions where n is a multiple of 3 corresponding to each group of three subintervals.

Let,

$$h = \frac{b-a}{n}$$

$$\begin{aligned} \int_a^b f(x) dx &= \text{Area}(\text{para}(x_0, y_0, y_1, y_2, y_3, h_3)) + \text{Area}(\text{para}(x_3, y_3, y_4, y_5, y_6, h_6)) + \dots + \text{Area}(\text{para}(x_{n-3}, y_{n-3}, y_{n-2}, y_{n-1}, y_n, h_n)) \\ &= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) + \frac{3h}{8} (y_3 + 3y_4 + 3y_5 + y_6) + \frac{3h}{8} (y_6 + 3y_7 + 3y_8 + y_9 + \dots + y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n) \end{aligned}$$

$$= \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_4 + y_7 + \dots + y_{n-1}) + 3(y_2 + y_5 + y_8 + \dots + y_{n-2}) + 2(y_3 + y_6 + y_9 + \dots + y_{n-3})]$$

(Practical)

* Evaluate the definite integration by using Simpson's $\frac{3}{8}$ rule.

Soln

Given that

$$\int_0^1 \frac{dx}{1+x^2}$$

$$f(x) = \frac{1}{1+x^2}$$

$$a=0, b=1, n=6$$

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
$f(x)$	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

* By Simpson's $\frac{3}{8}$:

$$\int_0^1 \frac{dx}{1+x^2} = \frac{3h}{8} \left[(y_0 + y_6) + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3) \right]$$

$$= \frac{3 \times \frac{1}{6}}{8} \left[\left(1 + \frac{1}{2}\right) + 3\left(\frac{36}{37} + \frac{9}{13}\right) + 3\left(\frac{9}{10} + \frac{36}{61}\right) + 2\left(\frac{4}{5}\right) \right]$$

$$= \frac{1}{16} \left[\frac{3}{2} + 3(1.66) + 3(1.49) + \frac{8}{5} \right]$$

$$= \frac{1}{16} \left[\frac{3}{2} + \frac{8}{5} + 4.98 + 4.47 \right]$$

$$= \frac{1}{16} [3.1 + 9.45]$$

$$= 0.7843$$